Developing Low Latency Applications in the Cloud with AI and ML

This article looks at the key considerations that are critical to the success of delivering low latency applications in the cloud. It also outlines how to build an interactive low latency application.

"Productivity is never an accident. It is always the result of a commitment to excellence, intelligent planning, and focused effort," said Paul J. Meyer, motivational speaker.

This was meant in the context of humans. With the evolution of technology and machines taking over mundane repetitive tasks from humans, this statement applies to machines as well today. Acceleration in the pace of change has led users to believe it's about time technology delivers instant response to requests, especially in the interactive world of entertainment (such as online video games like 'World of Warcraft' or 'Fortress', where hundreds of users participate simultaneously in voice conversations).

Edge computing has been around a long time; however, what it means to different industries and organisations has evolved over time differently. Cloud native computing has brought yet another meaning to the edge as a resource belonging to cloud service providers that can be leveraged on demand. Persistent efforts in the field of artificial intelligence (AI) over the last 50+ years have made it evolve from a scalable efficiency model to a scalable learning model by combining it with machine learning (ML).

The Internet of Things (IoT) has led to yet another increase in the demand for edge computing. The massive data that connected devices create by

communicating with each other over the Internet needs to be processed efficiently in order to be used properly. This has led to demand for low latency applications in the cloud.

Business considerations of low latency applications

While designing an application, business considerations must be the topmost priority in ensuring successful deployment of the solution. Here are the key business considerations that are critical to the success of delivering low latency applications.

- *Localised real-time data processing* will be efficient and deliver faster data driven decisions with the use of AI and ML.
- *Security at the edge*, as well as data encryption at rest and in transit, along with site key management for private and public key encryption will enable always encrypted communication.
- *Low operational costs with agility*, as bandwidth and throughput concerns typically are addressed with solutions that cost significant resources financially or otherwise.
- *Optimal affordable storage at the edge* and also at the hub as connected devices generate large amounts of data on a daily basis. Data analysis must be optimised at every point.

Technical considerations of low latency applications

A lack of understanding or planning of technical considerations leads to the vast majority of project failures. Let's look at some critical technical considerations that are important in the design and development of low latency applications.

- *Infrastructure deployment and configuration management* for the edge can be challenging. Hardware and software stack decisions must be made carefully ensuring the deployed architecture meets the expected performance considerations.
- *Edge network visibility* must be built beyond real-time traffic monitoring to include predictive and proactive analytics, leveraging AI and ML to create insights into edge performance.
- *Connectivity at the edge* for set application configurations should be designed for latency, bandwidth, throughput, and quality of service